

Anglo-Chinese Junior College
H2 Mathematics 9740
2009 JC 2 PRELIM Marking Scheme

Paper 2:

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Equation of line AB: $\mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ -4 \\ 1 \end{pmatrix}$ or $\mathbf{r} = \begin{pmatrix} 4 \\ -2 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ -4 \\ 1 \end{pmatrix}$

C is on the line AB: $\overrightarrow{OC} = \begin{pmatrix} 1+3\lambda \\ 2-4\lambda \\ 1+\lambda \end{pmatrix}$

Let $\theta = \angle AOC = \angle BOC$

$$\cos \theta = \frac{\overrightarrow{OA} \cdot \overrightarrow{OC}}{|\overrightarrow{OA}| |\overrightarrow{OC}|} = \frac{\overrightarrow{OB} \cdot \overrightarrow{OC}}{|\overrightarrow{OB}| |\overrightarrow{OC}|}$$

Hence, $\frac{\overrightarrow{OA} \cdot \overrightarrow{OC}}{|\overrightarrow{OA}|} = \frac{\overrightarrow{OB} \cdot \overrightarrow{OC}}{|\overrightarrow{OB}|}$

$$\frac{\begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 1+3\lambda \\ 2-4\lambda \\ 1+\lambda \end{pmatrix}}{\left| \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} \right|} = \frac{\begin{pmatrix} 4 \\ -2 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} 1+3\lambda \\ 2-4\lambda \\ 1+\lambda \end{pmatrix}}{2 \left| \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} \right|}$$

$$\frac{(1+3\lambda) + 2(2-4\lambda) + (1+\lambda)}{\sqrt{6}} = \frac{2[2(1+3\lambda) - (2-4\lambda) + (1+\lambda)]}{2\sqrt{6}}$$

$$(3-8+1)\lambda + (1+4+1) = (6+4+1)\lambda + (2-2+1)$$

$$15\lambda = 5$$

$$\lambda = \frac{1}{3}$$

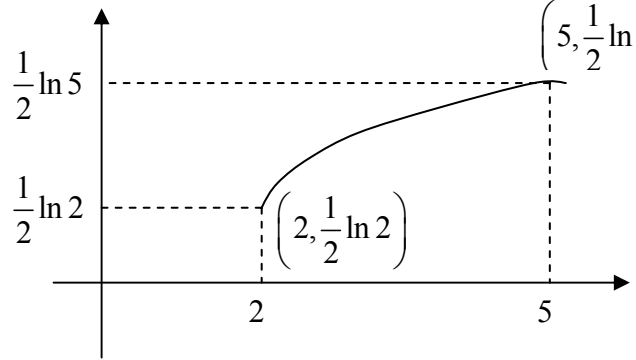
$$\overrightarrow{OC} = \frac{2}{3} \begin{pmatrix} 3 \\ 1 \\ 2 \end{pmatrix}$$

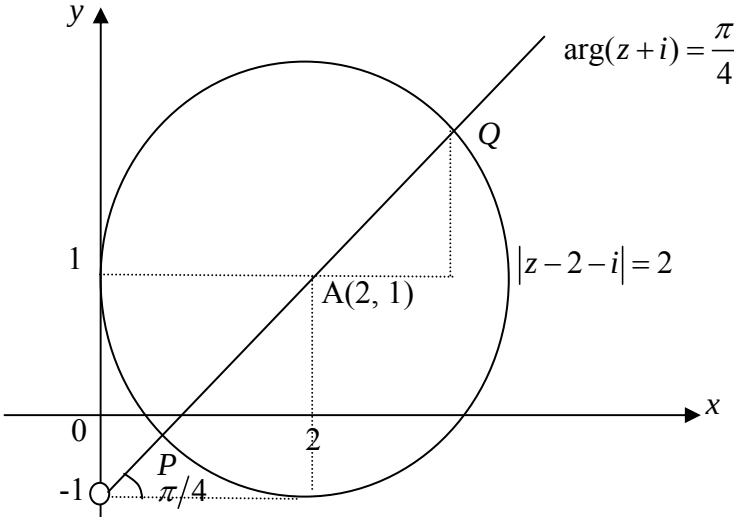
$$\overrightarrow{AC} = \overrightarrow{OC} - \overrightarrow{OA} = \frac{1}{3} \begin{pmatrix} 3 \\ -4 \\ 1 \end{pmatrix}$$

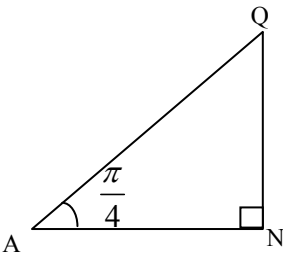
$$\overrightarrow{CB} = \overrightarrow{OB} - \overrightarrow{OC} = \frac{2}{3} \begin{pmatrix} 3 \\ -4 \\ 1 \end{pmatrix}$$

$$\frac{AC}{CB} = \frac{1}{2}$$

$$AC : CB = 1 : 2$$

2 (i)	$(2-ax)^{-b} = 2^{-b} \left(1 - \frac{ax}{2}\right)^{-b}$ $= 2^{-b} \left(1 + (-b)\left(-\frac{ax}{2}\right) + \frac{(-b)(-b-1)}{2!} \left(-\frac{ax}{2}\right)^2 + \dots\right)$ $= 2^{-b} \left(1 + \frac{ab}{2}x + \frac{a^2b(b+1)}{8}x^2 + \dots\right)$ Comparing constant term: $2^{-b} = \frac{1}{8} \Rightarrow b = 3$ Comparing coeff of x: $\frac{1}{8} \left(\frac{ab}{2}\right) = \frac{9}{16} \Rightarrow a = 3$ Comparing coeff of x^2: $\therefore c = \frac{1}{8} \left(\frac{a^2b(b+1)}{8}\right) = \frac{27}{16}$	
2 (ii)	$\text{coefficient of } x^r = 2^{-b} \frac{(-b)(-b-1)\dots(-b-r+1)}{r!} \left(-\frac{a}{2}\right)^r$ $= \frac{1}{8} \frac{(-3)(-4)\dots(-r-2)}{r!} \left(-\frac{3}{2}\right)^r$ $= \frac{1}{8} (-1)^{2r} \frac{(3)(4)\dots(r+2)}{r!} \left(\frac{3}{2}\right)^r$ $= \frac{1}{8} \frac{(r+1)(r+2)}{2} \left(\frac{3}{2}\right)^r$ $= \frac{1}{16} (r+1)(r+2) \left(\frac{3}{2}\right)^r$ $\therefore p = \frac{1}{16}, q = \frac{3}{2}$	
3 (i)		
3 (ii)	Required area $= \int_2^5 y \, dx$ $= \int_{\sqrt{2}}^{\sqrt{5}} (\ln t) 2t \, dt$ $= 2 \left\{ \left[\frac{t^2}{2} \ln t \right]_{\sqrt{2}}^{\sqrt{5}} - \int_{\sqrt{2}}^{\sqrt{5}} \frac{t^2}{2} \cdot \frac{1}{t} \, dt \right\}$ $= \frac{5}{2} \ln 5 - \ln 2 - \frac{3}{2}$ Therefore, $\alpha = \frac{5}{2}, \beta = -1, \gamma = -\frac{3}{2}$	

3 (iii)	Required volume = $\pi \left(\frac{1}{2} \ln 5 \right)^2 5 - \pi \left(\frac{1}{2} \ln 2 \right)^2 2 - \pi \int_2^5 (\ln \sqrt{x})^2 dx$ $= 5.75 \text{ square units}$	
4 (a)	$z = 1+i = \sqrt{1^2+1^2} = \sqrt{2}$, $w = \sqrt{3}+i = \sqrt{(\sqrt{3})^2+1^2} = 2$ $\arg(z) = \arg(1+i) = \frac{\pi}{4}$, $\arg(w) = \arg(\sqrt{3}+i) = \frac{\pi}{6}$ $\left \frac{z}{w} \right = \frac{ z }{ w } = \frac{\sqrt{2}}{2}$ $\arg\left(\frac{z}{w}\right) = \arg(z) - \arg(w)$ $\arg\left(\frac{z}{w}\right) = \frac{\pi}{4} - \frac{\pi}{6} = \frac{\pi}{12}$ $\frac{z}{w} = \frac{1+i}{\sqrt{3}+i} = \frac{1+i}{\sqrt{3}+i} \times \frac{\sqrt{3}-i}{\sqrt{3}-i} = \frac{1}{4} [(\sqrt{3}+1) + (\sqrt{3}-1)i]$ $\frac{z}{w} = \frac{\sqrt{2}}{2} \left(\cos \frac{\pi}{12} + i \sin \frac{\pi}{12} \right) = \frac{1}{4} [(\sqrt{3}+1) + (\sqrt{3}-1)i]$ Equating real parts: $\frac{\sqrt{2}}{2} \cos \frac{\pi}{12} = \frac{1}{4}(\sqrt{3}+1)$ Equating imaginary parts: $\frac{\sqrt{2}}{2} \sin \frac{\pi}{12} = \frac{1}{4}(\sqrt{3}-1)$ $\tan \frac{\pi}{12} = \frac{\sqrt{3}-1}{\sqrt{3}+1} \times \frac{\sqrt{3}-1}{\sqrt{3}-1} = 2 - \sqrt{3}$ $= \frac{\sqrt{3}-1}{\sqrt{3}+1} \times \frac{\sqrt{3}-1}{\sqrt{3}-1}$ $= 2 - \sqrt{3}$	
4 (b)	 The diagram shows a complex plane with x and y axes. A circle is centered at point A(2, 1) with a radius of 2, as indicated by the equation $z-2-i =2$. The circle passes through the point (0, 1) on the y-axis and (2, -1) on the x-axis. A ray originates from the origin and passes through point Q on the circle in the first quadrant. This ray is labeled with the argument $\arg(z+i) = \frac{\pi}{4}$. Another point P is marked on the negative y-axis at (0, -1), and an angle of $\pi/4$ is shown between the negative y-axis and a horizontal dashed line from P.	

	$AQ = 2$ $AN = AQ \cos \frac{\pi}{4} = 2 \left(\frac{\sqrt{2}}{2} \right) = \sqrt{2}$ $NQ = AQ \sin \frac{\pi}{4} = 2 \left(\frac{\sqrt{2}}{2} \right) = \sqrt{2}$ x-coordinate of Q: $2 + \sqrt{2}$ y-coordinate of Q: $1 + \sqrt{2}$ x-coordinate of P: $2 - \sqrt{2}$ y-coordinate of P: $1 - \sqrt{2}$ Coordinates of Q: $(2 + \sqrt{2}, 1 + \sqrt{2})$ Coordinates of P: $(2 - \sqrt{2}, 1 - \sqrt{2})$ Hence $z = (2 + \sqrt{2}) + (1 + \sqrt{2})i$ or $z = (2 - \sqrt{2}) + (1 - \sqrt{2})i$ 	
5	The number of seating arrangements = $4! = 24$ Number of ways = $3! \times 2! = 12$ Number of ways = $4! \times 2! \times 6 = 288$	
6 (i)	Let X be the random variable for the number of vehicles arriving in 2 minutes $X \sim \text{Po}(20)$ $P(X = 18) = 0.0844$	
6 (ii)	Vehicles: $180 \text{ min} = 3 \text{ hr} = 1800$ Variance = $\text{Var}(0.7X) = 0.7 \times 0.7 \times 1800 = 882$	
6 (iii)	Let C be the random variable for the number of cars $C \sim B(50, 0.7)$ Since $np = 50 \times 0.7 = 35 > 5$ and $nq = 50 \times 0.3 = 15 > 5$ Using Normal approximation to Binomial $C \sim N(35, 35(0.3))$ i.e $N(35, 10.5)$ $P(C \geq 30) = P(C > 29.5)$ (with cc) $= 0.955$	
7 (i)	Stratified sampling is more representative of the entire population. Random sampling can't ensure that both genders are represented appropriately . <i>(A random sample might produce all boys and their eating preferences may be different from girls)</i>	
7 (ii)	Biased or not random. The population of leaves beyond the student's reach has been excluded .	
8	Let W be the random variable for the number of matches of tennis won by Wayne $X \sim B(120, \frac{1}{2})$ $P(W \geq 70) = 0.0412$ Probability that Wayne won 70 or more matches is only 0.0412.	

	It does not seem likely that Henry and Wayne are equally skilled players.	
9 (i)	$n = 50$, large using Central Limit theorem $\bar{X} \sim N(20\ 000, \frac{2200^2}{50})$ $P(\bar{X} - 20\ 000 < 800) = P(19\ 200 \leq \bar{X} \leq 20\ 800) = 0.990$	
9 (ii)	Let Y be the random variable for number of samples with sample mean within \$800 of the population mean $Y \sim B(5, 0.989868)$ $P(Y = 3) = 9.96 \times 10^{-4}$	
9 (c)	Assume : $P(\text{success}) = 0.989868$ is a constant for each trial	
10 (i)	Let X be the random variable for the length of a pickle $X \sim N(9, 0.8^2)$ (a) $P(X < 8) = 0.106$ (b) $P(X > 10.5) = 0.0304$	
10 (ii)	Since 10.6% of the pickles are discarded because they are too short , it is impossible to reduce the total rejects to 5%	
10 (iii)	Let L be the random variable for the number of pickles which are too long $L \sim B(100, 0.0304)$ $P(L > 8) = 1 - P(L \leq 8)$ Since $n = 100$ large, $p = 0.0304 < 0.1$ and $np = 3.04 < 5$ Using Poisson approximation to Binomial $L \sim \text{Po}(3.04)$ $P(L > 8) = 0.00413$	
10 (iv)	$P(\text{both pickles to meet his guaranteed standard} \mid X_1 + X_2 < 16 \text{ cm})$ $= \frac{P(\text{both pickles fail his g'teed standard} \ \& \ X_1 + X_2 < 16)}{P(X_1 + X_2 < 16)}$ $= \frac{(P(X < 8))^2}{P(X_1 + X_2 < 16)} \dots\dots(***)$ $= \frac{(0.1056498)^2}{0.03854989} = 0.290$	
11 (i)	To test $H_0: \mu = 3.6$ against $H_1: \mu < 3.6$ at 1 % level of significance	
11(ii)	Under H_0 , $\bar{X} \sim N(3.6, \frac{1.2^2}{n})$ Test statistics: $Z = \frac{\bar{X} - \mu_0}{\frac{\sigma}{\sqrt{n}}} = \frac{3.5 - 3.6}{\frac{1.2}{\sqrt{n}}} = \frac{-0.1}{\frac{1.2}{\sqrt{n}}}$	

	$p\text{-value} = P(\bar{X} < 3.5) = P\left(Z < \frac{-0.1}{\frac{1.2}{\sqrt{n}}}\right) < 0.01 = P(Z < -2.32635),$ Reject H_0, hence $\frac{-0.1}{\frac{1.2}{\sqrt{n}}} < -2.32635$. $n > (2.32635 \times 12)^2 = 779.314$ Hence minimum $n = 780$	
11 (iii)	Carry out same test as above at 5% level of significance $n = 15, \quad \sum x = 47 \quad \text{Hence } \bar{x} = \frac{\sum x}{n} = \frac{47}{15} = 3.13$ $\sum (x-1) = \sum x - \sum 1 = 47 - 1(15) = 32$ $s^2 = \frac{1}{n-1} \left[\sum (x-1)^2 - \frac{(\sum (x-1))^2}{n} \right] = \frac{1}{14} \left[90 - \frac{(32)^2}{15} \right] = 1.55238$ using the t-distribution $\text{Test statistics } t = \frac{\bar{X} - \mu_0}{\frac{s}{\sqrt{n}}} = \frac{3.1333 - 3.6}{\frac{\sqrt{1.55238}}{\sqrt{15}}} = -1.45$ $p\text{-value} = 0.0845 > 0.05$ Do not reject at 5% level of significance and conclude that there is insufficient evidence that the modification has decreased the mean diameter of the ball bearings at 5% level of significance.	
11 (iii)	If $\mu = 3.6$, in fact 100p% of tests using the same method will lead to wrongly rejecting $\mu = 3.6$ or p value is the probability of wrongly rejecting the hypothesis that $\mu = 3.6$	
12 (i)	$r = -0.952$ For the depths sampled the moisture content decreases approximately linearly with an increase in the depth of the sample. Since r is close to 1, this implies that the regression line of m on x and x on m are almost identical or close to each other	

12 (ii)	$m = -2.2048x + 83.583.$ When $m = 50$, $x = \frac{83.583 - 50}{2.2048} = 15.232 = 15.2$ Reliable since the value of m is within the range of the given data. Not extrapolating.	
12 (iii)	$r_{yx}^2 = (-0.904)^2 = bd = -2.33 d$ Hence $d = -0.351$ Hence regression line of x on y is $x = -0.351y + C$ Since $\bar{x} = 17.5$ and $\bar{y} = -2.33(17.5) + 50 = 9.225$ $(17.5, 9.225)$ also falls on the line of regression of x on y Hence $C = 20.7$ Regression line of x on y is $x = -0.351y + 20.7$	
13	Let U and R be the events that United and Rover won the match respectively. P(match is still undecided after 1 round) $= P(R \cap U) + P(R' \cap U') = 0.8 \times 0.9 + 0.2 \times 0.1 = 0.74 = \frac{37}{50}$	
13 (i)	P(United won in less than 3 rounds United scores a goal in the first round) $= \frac{P(\text{United won in less than 3 rounds and United scores a goal in the first round})}{P(\text{United scores a goal in the first round})}$ $= \frac{0.8 \times 0.1 + 0.8 \times 0.9 \times 0.8 \times 0.1}{0.8} \dots\dots\dots (*)$ $= 0.172$	
13 (ii)	Let X be the random variable for the number of rounds played P(match is decided in at most n rounds) > 0.98 $P(X \leq n) > 0.98$ $P(X = 1) + P(X = 2) + P(X = 3) + \dots\dots + P(X = n) > 0.98$ Hence $\frac{13}{50} + \frac{37}{50} \left(\frac{13}{50}\right) + \left(\frac{37}{50}\right)^2 \frac{13}{50} + \dots\dots + \left(\frac{37}{50}\right)^{n-1} \frac{13}{50} > 0.98 \dots\dots (**)$ $\frac{13}{50} \left[1 + \frac{37}{50} + \left(\frac{37}{50}\right)^2 + \dots\dots + \left(\frac{37}{50}\right)^{n-1} \right] > 0.98$ $\left(\frac{37}{50}\right)^n < 0.02$ Hence $n > \frac{\ln 0.02}{\ln \frac{37}{50}} = 12.992$ Least $n = 13$	